Analytical methods for timing aspects of the transport of CBR services over ATM

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The transport of constant bit rate (CBR) services using asynchronous transfer mode (ATM) has been widely discussed in international standards fora. To date, there has been little discussion of the effects of 'waiting time jitter' on performance and this paper presents an analysis and set of simulations of this phenomenon. It shows that systems which choose to use a phase-locked loop (PLL) to recover the CBR service from the ATM cell stream will encounter this phenomenon and dealing with it will place very stringent conditions on the design of the PLL. By comparison, other techniques which use a network clock appear simple and robust.

1. Introduction

Constant bit rate (CBR) services are an important class of service and their effective transport using asynchronous transfer mode (ATM) is critical to the success of ATM as a universal transfer mode. This paper sets out a means by which the impairments caused by ATM on CBR services can be described and quantified.

Any CBR service can be considered in two parts, firstly the data stream which is the binary information, and secondly a clock signal. The clock signal is used by a receiving device to assign each bit or piece of information to its correct temporal position with respect to the other bits or pieces of information. In principle, it is this clock signal that is the defining feature of a CBR service. For the purposes of this paper, only the clock signal is referred to as this is the essential part of the CBR service within the context of this paper.

2. Methods of describing clock signals

A clock signal is a regular signal which carries time information. The critical information is therefore when a signal reference mark occurs (e.g. a zero crossing in a physical clock waveform). This is illustrated in Fig 1. Several terms are used to describe the clock signal, all of which relate back to the time at which the clock signal pulses occur. The most common are time (measured normally in ns), unit intervals (UI, where 1 UI corresponds to one cycle of the clock signal), and phase (measured in radians where 1 UI = 2 π radians).

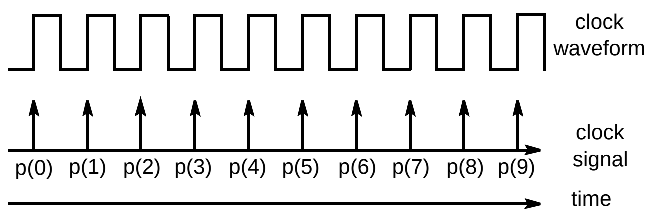


Fig 1 A clock signal.

The quality of a clock signal depends on the accuracy of the timing information. Impairments in this information give errors in the time when the clock pulses occur as illustrated in Fig 2. Again several terms are used to describe this impairment, the most general of which is phase variation.

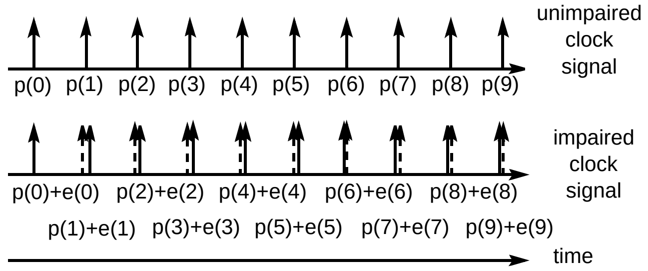


Fig 2 Phase variation on a clock signal.

Phase variation is simply the value of the error signal $e(n\tau)$, where τ is the clock period. This is often written as a continuous function of time $e(t)$ and can be measured in ns, UI, or radians. There are several parameters which can be used to characterise $e(t)$. The two most common are jitter and wander. Jitter is a measure of the higher frequency characteristics of the phase variation while wander is the lower frequency characteristics of phase variation. The 'dc' component (i.e. absolute phase drift or frequency offset) is sometimes considered independently and sometimes as wander.

3. Why phase variation is important

All practical physical interfaces require a good clock signal. The existing plesiochronous digital hierarchy (PDH) interfaces in telecommunications have specifications for their clock signals which, while different in some details, are all basically similar. These are summarised in Table 1.

Table 1 Specification of phase variation.

CBR bit rate	Maximum frequency offset	Maximum section jitter (pk-pk)	Maximum end-to-end jitter (pk-pk)	Jitter measurement HPF
1544 kbit/s	+/- 150 ppm	1.5 UI	5 UI	10 Hz
2048 kbit/s	+/- 50 ppm	0.4 UI	1.5 UI	20 Hz
6132 kbit/s	+/- 30 ppm	1.5 UI	5 UI	10 Hz
8448 kbit/s	+/- 30 ppm	0.4 UI	1.5 UI	20 Hz
34368 kbit/s	+/- 20 ppm	0.4UI	1.5 UI	100 Hz
44736 kbit/s	+/- 20 ppm	1.5 UI	5 UI	10 Hz
139264 kbit/s	+/- 15 ppm	0.4 UI	1.5 UI	200 Hz

The consequence of not meeting these requirements is profound. All the signal recovery circuits in existing equipment are designed to meet these limits and exceeding them will inevitably lead to misalignment in the equipment. This will give random, spurious error bursts at frequent intervals, the cause of which will be almost impossible to trace. All existing network practice assumes that equipment is checked to be consistent with these limits before it is commissioned into the network. Any relaxation of these limits would be disastrous for the operation of the existing network and, at least for these rates, these jitter limits must be met.

One of the advantages of ATM is that is will support any arbitrary CBR however, in this case the jitter and wander limits may not be so well defined. Even so, jitter must be kept under strict control if signal recovery circuits are going to work reliably. While in these cases, the limits may be different, they must exist and be met.

In summary, while the subject of phase variation is given little profile, largely because limits have been strictly adhered to in the past, there are few network impairments which are more able to severely disrupt the operation of the existing network and in such a way that the cause cannot be easily located.

4. System model and method of analysis

The general configuration of a CBR service transported on ATM is illustrated in Fig 3.

4.1 *ATM impairments — justification effects and cell delay variation*

The CBR signal enters the ATM network and the first function this must perform is the conversion of the regular clock of the CBR signal into a cell stream. The nature of the cell stream has the effect of 'quantising' both the phase of the signal and the time of the signal. The phase can only be sent in discrete 'quanta', i.e. cells, and a cell can only be sent when the cell occurs in the ATM cell stream.

This dual quantisation has the effect of setting up a pattern of when cells are filled and when they are not. This is normally called a justification pattern. This phenomenon was first recognised with PDH multiplexing systems and the control of these patterns was the primary design objective of the PDH frame structures. The justification

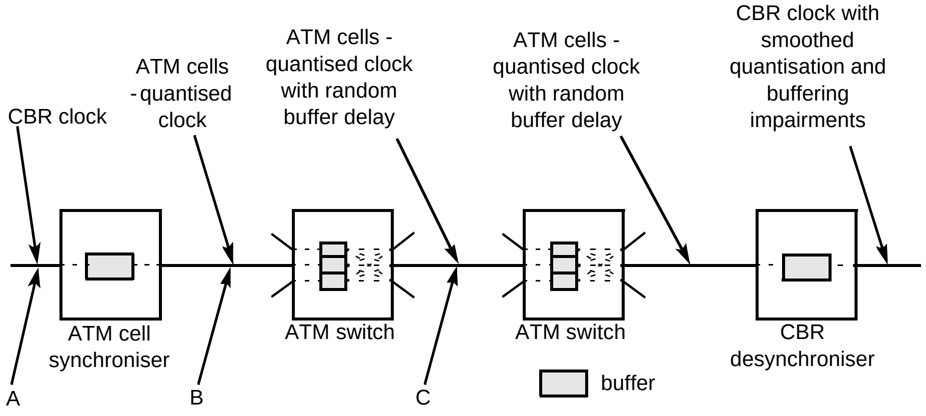


Fig 3 General reference connection for CBR services.

patterns are extremely complex with a strong ‘fractal’ nature (this can be clearly seen from the simulations later in the paper). Much work was done in the early days of PDH multiplexing in order to understand these patterns; however, a lot of the expertise has since disappeared and in ATM standardisation, the issue has been almost entirely ignored. Much of this paper is about re-applying this understanding to ATM.

The second effect is when the ATM cell stream enters an ATM switch. The output port of the switch statistically multiplexes the CBR cell stream with cells from other connections and there is a significant probability that when a cell from the CBR connection arrives at the switch, the first output cell is already occupied by a cell from another connection. The CBR connection cell must be buffered until there is a free output cell available. This gives rise to ‘cell delay variation’ (CDV) . The buffer effectively delays the cell’s transit through the switch and, since the extent of buffering depends of the random statistics of the multiplex, the cell delay variation will also be statistical.

To date, most of the modelling of CBR services has centred on CDV and has made explicit assumptions about the stochastic processes producing the CDV. Justification patterns can be significant under certain conditions and the transient effects of CDV when loading changes rapidly may well be more significant than ‘steady state’ stochastic processes producing CDV.

The justification and cell delay variation effects are illustrated in Fig 4.

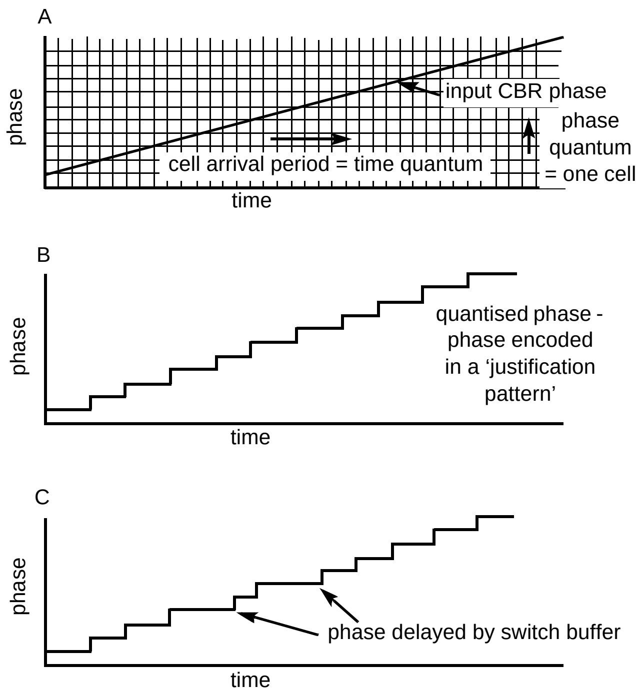


Fig 4 Effects of phase quantisation and switch buffer delay.

4.2 *Desynchronising — recovering the CBR clock*

The job of the desynchronising function is to recreate the CBR clock from the cell stream. There are broadly two classes of desynchroniser:

- use of a phase-locked loop (PLL) which tracks and averages the arrival of the phase quanta (i.e. cells) over a long period of time and uses this moving average to generate the CBR clock (although this is sometimes called an ‘adaptive technique’, for a number of reasons, this term can be misleading as it does not capture the essential control-loop nature and also tends to imply non-linearity which need not be the case),
- the data is sent out with a locally generated, independent clock and any difference that might occur between that clock and the original CBR clock is taken up with a ‘slip buffer’ which can insert or delete bits from the data stream if the need arises.

The PLL technique has the advantage that it will guarantee that every bit on the input CBR interface is faithfully reproduced on the output CBR interface. Its disadvantage is that the clock cannot be faithfully reproduced and timing impairments are inevitable. The design of the PLL must ensure that these impairments are within acceptable limits.

Conversely, the slip buffer technique cannot guarantee the bit level integrity of the data unless the clocks are original CBR clocks and desynchonising clocks are synchronous with each other. Even if this is the case under normal circumstances, it may not be the case under some network failure conditions and this may not be the case across operator boundaries.

Both techniques therefore have advantages and disadvantages. Both PDH and SDH (synchronous digital hierarchy) generally use the PLL mechanism while the 64 kbit/s network uses the slip buffer mechanism, and so both techniques are well established in the telecommunications network. Having said that, the PDH uses a maximum phase quantum of 1 UI and SDH a maximum of 24 UI (and more normally 8 UI). ATM has a basic quantum of 376 UI (i.e. 47 octets) and so the difficulties associated with the PLL, which increased by an order of magnitude moving from PDH to SDH, increase by another order of magnitude with ATM.

Three schemes for desynchronising CBR services have been proposed and are described in Mulvey, Kim and Reid [1]:

- for network synchronous CBR services (i.e. 64 kbit/s and *n* x 64 kbit/s) a simple slip buffer technique is used,

- for CBR services which are generated using a local clock (e.g. 2 Mbit/s and 1.5 Mbit/s) a pure PLL technique is used,

- alternatively, for CBR services generated using a local clock, a hybrid system is used where the CBR clock is measured at the input to the ATM network relative to the network clock and this information is sent in the cells to the desynchronising end, where it uses its network clock with this information to reconstruct the CBR clock — while this does have many of the features of the PLL techniques, it is still essentially a slip buffer technique.

In principle, the first and third (i.e. slip buffer) techniques work within the constraints of network synchronisation. The second (i.e. the PLL) is attractive from the point of view that it does not require reliable network synchronisation in order to work. This paper addresses the viability of the PLL solution and the conditions it must meet in order to work satisfactorily and reliably in the telecommunications network.

4.3 Modelling a PLL desynchroniser

There are two stages to modelling the PLL desynchroniser and quantifying its effects. Firstly, there is the desynchroniser itself which, despite being of proprietary design, will have general characteristics which can be universally modelled. Secondly, there is the way the effects are measured. The output will contain both jitter and wander, the cut-off between the two being distinguished by the pole of the measurement filter. While both excessive jitter and wander can cause significant operational problems, in general it is the jitter problems that will be the more widespread and so the analysis concentrates on the jitter. Having said that, in situations where wander is a problem, the effects of ATM are much more insidious — they cannot ever be overcome by a PLL design.

Fig 5 illustrates the way the PLL desynchroniser and the jitter measurement are modelled.

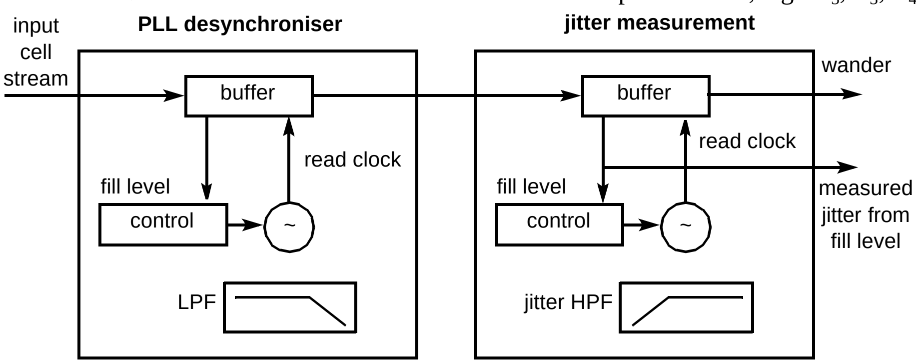


Fig 5 Model of PLL desynchroniser and jitter measurement.

In general, the easiest model of the PLL desynchroniser and the jitter measurement filter is as single pole filters. While this will not be exactly accurate, the constraints of closed loop stability require that any design will not be far removed from a single pole filter. The PLL desynchroniser is a low-pass filter while the jitter measurement filter is a high-pass filter. The general shape of the phase waveforms at each point is illustrated in Fig 6.

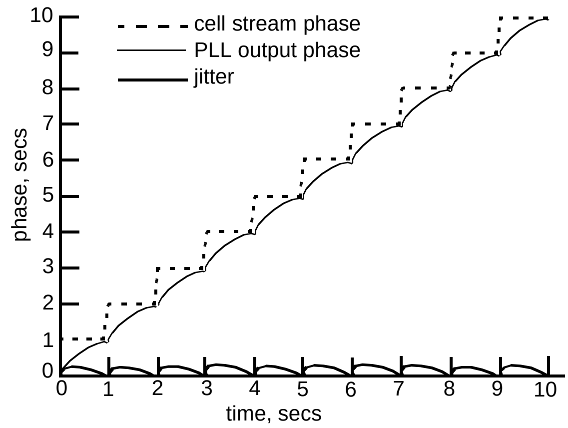


Fig 6 Phase waveforms.

5. Analytical understanding of waiting time jitter

The characteristics of waiting time jitter are very complex. Although a simple analysis of waiting time jitter is not realistic, it is possible to infer a great deal about the nature of the jitter for a given set of conditions. Figure 7 gives an impression of the nature of waiting time jitter (it is one of the simulated examples).

Figure 7 shows that the waiting time jitter has clear peaks and that the distribution of these peaks across the range of possible CBR rates is neither uniform nor random — it is ‘fractal’. The peaks are actually associated with rational fractions. The peak in the centre is associated with $^1/_2$ while the large peaks surrounding it are associated with the other simple fractions, e.g. $^1/_3$, $^2/_3$, $^1/_4$, $^3/_4$, etc. On closer inspection, it can be seen that the peaks surround the actual fractional value. At the precise fractional value, the waiting time jitter is very low.

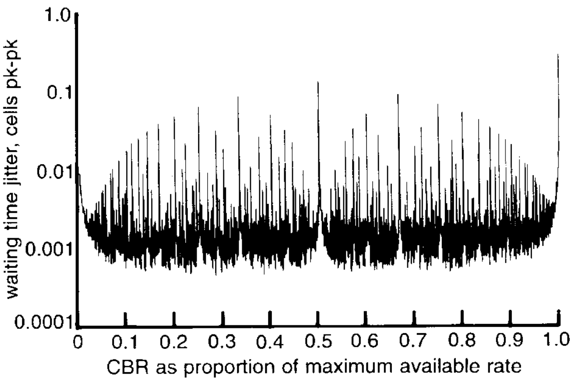


Fig 7 Characteristics of waiting time jitter.

This gives the clue to the mechanism that produces the peaks. When the CBR rate is very close to a simple fraction of the available rate, the cells that are used for the CBR settle into a fixed pattern. For example, at a CBR rate of around $^2/_3$, two cells in three will normally be filled with the CBR signal. However, very occasionally, the pattern will reset if the CBR rate is not exactly $^2/_3$. The period between these resets can be long and it is this time that gives rise to the term 'waiting time jitter'. This is the time the desynchroniser must wait to receive the phase information from the cell stream. The used cells can now be considered as the superposition of a fixed pattern associated with the rational fraction and periodic resets. The amplitude of the reset is determined by the denominator of the rational fraction. The amplitude of the reset associated with $^2/_3$ is $^1/_3$ of a cell. If the actual rate is just above the rational fraction, the resets are positive, and if it is below the fraction, the resets are negative. (In Fig 7 the peaks are attenuated to 25% of their reset value by the combination of the desynchroniser (10 Hz) and the measurement filter (20 Hz) — in this case, the peak associated with $^2/_3$ is measured as $^1/_{12}$ of a cell.) The jitter at the actual rational fraction is that associated with the fixed pattern, and the more complex the pattern (i.e. the larger the denominator), the larger this low-level jitter.

It can also be seen that the peaks have a width. This width is determined by the relative frequency of the resets and the pole of the high-pass measurement filter (in the case of Fig 7 this is 20 Hz). If the frequency of the resets is below this pole, they are then passed through as single isolated phase transients; if the frequency is higher, then the transients will start to coincide and effectively attenuate each other.

In order for the waiting time jitter to be understood, the following must be quantified:

- the basic attenuation of isolated transients associated with the two filters,
- the width of the peaks,
- the number of peaks (broadly the inverse of the peak width),
- the probability of encountering a high peak under normal operation.

In order to analyse this, consider a signal close to a peak which has a reset amplitude of A, a period between resets of T, a desynchroniser with a primary closed loop pole at ω_L, and a jitter measurement filter with a pole of ω_H. This analysis is illustrated in Fig 8.

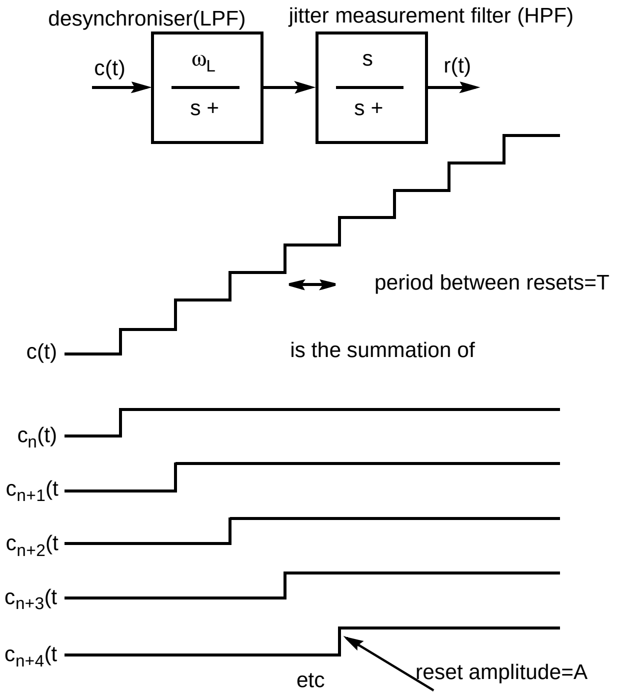


Fig 8 Analysis of waiting time jitter.

Firstly the response from one of the individual reset transients is examined:

$$R_n(s) = \frac{A}{s} \cdot \frac{s'}{s + \varpi_H} \cdot \frac{\varpi_L}{s + \varpi_L}$$

$$= \frac{A\varpi_L}{(\varpi_H - \varpi_L)} \left[\frac{1}{s + \varpi_L} - \frac{1}{s + \varpi_H} \right]$$

$$\Rightarrow r_n(t) = \frac{A\varpi_L}{\varpi_H - \varpi_L} \left[e^{-\varpi_L(t - nT)} - e^{-\varpi_H(t - nT)} \right] U(t - nT)$$

where T is the time interval between the transient responses.

The steady-state response is the sum of all the previous transients and is a geometric series and so the summation can be evaluated:

$$r(t) = \sum_{i=0}^{-\infty} r_i(t) \qquad nT \leq t < (n+1)T$$

$$= \frac{A\varpi_L}{(\varpi_H - \varpi_L)}\left[\frac{e^{-\varpi_L(t-nT)}}{1-e^{-\varpi_L T}} - \frac{e^{-\varpi_H(t-nT)}}{1-e^{-\varpi_H T}}\right]$$

$$nT \leq t < (n+1)T$$

This equation gives the absolute value of the offset in the buffer of the measurement device and is a combination of static phase offset (which should be proportional to the bit of the constant bit rate encoded in the phase transients) and waiting time jitter. The static phase offset can be evaluated by integrating and averaging the waveform over a cycle:

$$\mu = \frac{1}{T}\int_{nT}^{(n+1)T} r(t)dt \qquad nT \leq t < (n+1)T$$

$$= \frac{A\varpi_L}{T(\varpi_H - \varpi_L)}$$

$$\left[\frac{-\frac{1}{\varpi_L}e^{-\varpi_L(t-nT)}}{1-e^{-\varpi_L T}} - \frac{-\frac{1}{\varpi_H}e^{-\varpi_H(t-nT)}}{1-e^{-\varpi_H T}}\right]_0^T$$

$$= \frac{A}{\omega_H T}$$

which is indeed the static phase offset, proportional to the transient height, the frequency of the transients, and inversely proportional to the pole of the measurement filter.

The peak-to-peak jitter is the difference between the minimum of the waveform and the maximum of the waveform. In this particular example, it can be seen from the waveform that the minimum will occur at $t = nT$ and the maximum when $r'(t) = 0$.

The minimum is:

$$r_{min} = \frac{A\varpi_L}{(\varpi_H - \varpi_L)}\left[\frac{1}{1-e^{-\varpi_L T}} - \frac{1}{1-e^{-\varpi_H T}}\right]$$

The maximum is calculated as follows:

$$r'(t) = \frac{A\varpi_L}{(\varpi_H - \varpi_L)}\left[\frac{-\varpi_L e^{-\varpi_L(t-nT)}}{1-e^{-\varpi_L T}} - \frac{-\varpi_H e^{-\varpi_H(t-nT)}}{1-e^{-\varpi_H T}}\right]$$

$$nT \leq t \leq (n+1)T$$

$$= 0$$

$$\Rightarrow \frac{-\varpi_L e^{-\varpi_L(t-nT)}}{1-e^{-\varpi_L T}} = \frac{-\varpi_H e^{-\varpi_H(t-nT)}}{1-e^{-\varpi_H T}}$$

$$\Rightarrow t_{max} - nT = \frac{1}{\varpi_H - \varpi_L} 1n\left(\frac{\varpi_L(1-e^{-\varpi_H T})}{\varpi_H(1-e^{-\varpi_L T})}\right)$$

$$\Rightarrow r_{max} = \frac{A\varpi_L}{(\varpi_H - \varpi_L)}$$

$$\left[\frac{\left(\frac{\varpi_L(1-e^{-\varpi_H T})}{\varpi_H(1-e^{-\varpi_L T})}\right)^{-\frac{\varpi_L}{\varpi_H - \varpi_L}}}{1-e^{-\varpi_L T}} - \frac{\left(\frac{\varpi_L(1-e^{-\varpi_H T})}{\varpi_H(1-e^{-\varpi_L T})}\right)^{-\frac{\varpi_L}{\varpi_H - \varpi_L}}}{1-e^{-\varpi_L T}}\right]$$

The peak-to-peak jitter is therefore:

$$r_{pk-pk} = r_{max} - r_{min}$$

$$= \frac{A\varpi_L}{(\varpi_H - \varpi_L)}$$

$$\left[\frac{\left(\frac{\varpi_L(1-e^{-\varpi_H T})}{\varpi_H(1-e^{-\varpi_L T})}\right)^{-\frac{\varpi_L}{\varpi_H - \varpi_L}} - 1}{1-e^{-\varpi_L T}} - \frac{\left(\frac{\varpi_L(1-e^{-\varpi_H T})}{\varpi_H(1-e^{-\varpi_L T})}\right)^{-\frac{\varpi_L}{\varpi_H - \varpi_L}} - 1}{1-e^{-\varpi_L T}}\right]$$

In the situation where the period between the transients is significantly longer than the time constants of the filters, i.e. $T >> 1/\omega_L$ and $T >> 1/\omega_H$, then the above equation simplifies to:

$$r_{pk-pk} = r_{max} - r_{min}$$

$$= \frac{A\omega_L}{(\omega_H-\omega_L)}\left[\left(\frac{\omega_L}{\omega_H}\right)^{-\frac{\omega_L}{\omega_H-\omega_L}} - \left(\frac{\omega_L}{\omega_H}\right)^{-\frac{\omega_L}{\omega_H-\omega_L}}\right]$$

These last two equations can be used to characterise much of the waiting time jitter.

Figure 9 shows a plot of the general equation with dimensionless ratio $\omega_L T$ on the *x* axis and the dimensionless ratio ω_L/ω_H of the *y* axis (into the page). It can be seen that the general shape is controlled by $\omega_L T$ until it becomes roughly equal to eight times ω_L/ω_H (note the factor of eight can be derived from the above equations). Until this point, the attenuation is 0.125 $\omega_L T$ while after this point is becomes roughly equal to ω_L/ω_H. This is true so long as $\omega_L < \omega_H$ and, if this is not the case, there is almost no attenuation of the transient.

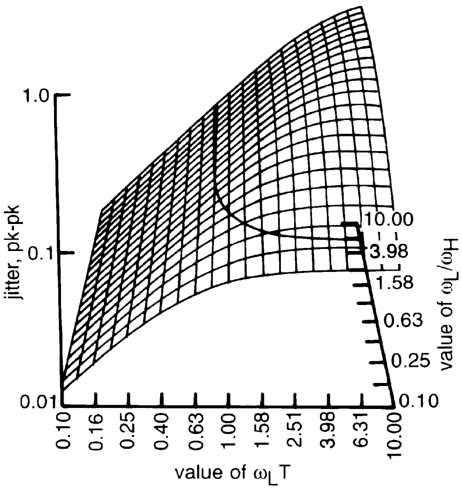


Fig 9 Plot showing attenuation of waiting time jitter peaks.

This also indicates the width of the peaks. For any given ω_L and ω_H, all the peaks in the waiting time jitter will have a flat top corresponding to the point when *T* is such that:

$$\omega_L/\omega_H = 0.125\ \omega_L T$$
$$T = 8 / \omega_H$$
$$T \approx 1/f_H$$

$$\text{width} \approx \frac{f_H}{\text{cell rate}}$$

where it is in the fraction of the maximum available rate and it will roll off from the flat top at 20 dB/decade. The number of peaks will be approximately the inverse of their width:

number of peaks $\approx 1/\text{width}$

The height of the peaks is controlled by $A\omega_L/\omega_H$ when $\omega_L \ll \omega_H$. For the largest peaks $A = 1$, and for the smallest peaks A is the inverse of the number of peaks (which must be of the same order as the denominator of the fractional rate):

smallest $A \approx$ width

Because the number of peaks of a given height is roughly the same as the size of the denominator:

$P(A) \approx \text{width}/A$

where $P(A)$ is the probability of an arbitrary CBR having a given value of A.

6. Results of system simulation

Figure 7 showed the results of a simulation of the effects of waiting time jitter on a CBR service with cells carried in a payload of 1920 kbit/s, a PLL desynchroniser pole of 10 Hz and a measurement filter of 20 Hz. Figure 10 shows the similar situation; however, this time the PLL desynchroniser pole has now been reduced to 1 Hz. By comparing Figs 7 and 10, it is possible to see that the only effect of this is to reduce the height of the peaks by a factor of ten and the overall shape of the peaks is unchanged.

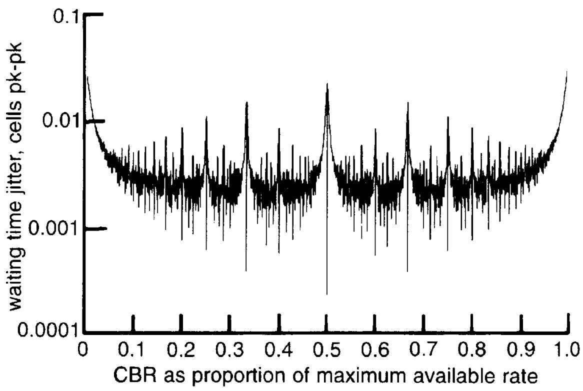


Fig 10 Cells in 1920 kbit/s with LPF = 1 Hz and HPF = 20 Hz.

If the cells are carried in a payload of 149760 kbit/s, i.e. that of the SDH VC-4, this has the effect of reducing the width of the peaks, increasing the number of the peaks, but leaving their height unchanged when compared to the case with cells in a 1920 kbit/s payload. This is illustrated in Fig 11.

The particular case of a 2048 kbit/s CBR service carried in cells which are themselves in an SDH VC-4, even when spread over the full range of +/ −50 ppm, does not appear to encounter a significant peak as illustrated in Fig 12. This clearly suggests that a more general conclusion drawn from practical experiments associated with this case will not be universally valid.

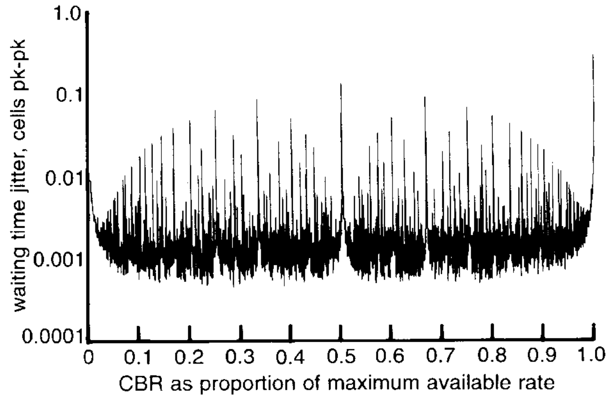


Fig 11 Cells in VC-4 with LPF = 10 Hz and HPF = 20 Hz.

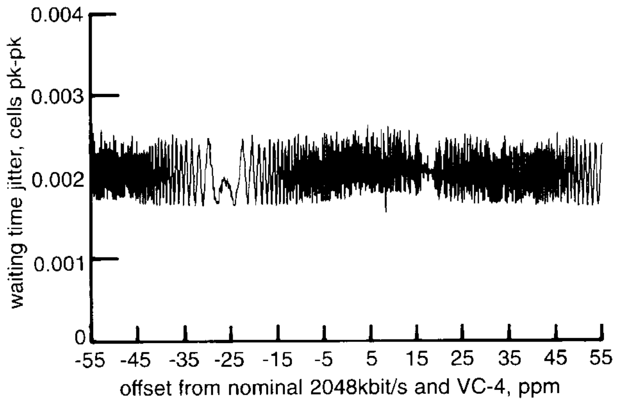


Fig 12 Detailed region around 2048 kbit/s in VC-4.

7. Further effects of cell delay variation

Any ATM multiplexer, crossconnect, or switch will multi-plex a number of ATM virtual channels together on an interface port. The hardware required in order to achieve this multiplex will have a buffer and the size characteristics of this buffer are discussed in Mulvey, Kim and Reid [1]. The existence of the buffer gives rise to a delay in the signal; however, this delay does not directly give rise to any timing impairments. Impairments arise when this delay changes with time, i.e. cell delay variation (CDV). The exact nature of the impairment will depend on the precise stochastic nature of the CDV.

An analytical solution will depend on the stochastic model assumed for the multiplex and many have been offered in published papers. Most of these have dealt with the absolute size of the delay which is important for sizing play-out buffers for the network synchronous and SRTS schemes. In order to analyse the effect on adaptive PLL-based schemes, the spectral nature of the CDV is also required. In addition, most analysis is based on a 'steady state' loading assumption on the switch [1] and does not address the condition of loading transients.

Transients are likely to be the worst case from the point of view of jitter. If the loading of a switch increases over a very short period of time in such a way that the CBR service experiences an almost instantaneous increase in delay, an adaptive PLL-type of desychroniser will see a single large step in phase associated with this loading increase. This will cause a jitter transient in exactly the same way as waiting time jitter. The size of this transient will depend on the number of cells of delay added in the buffer as a result of the load increase and the scaling factor of this effect which is the proportion of the CBR rate to the output cell rate, i.e.:

$$c(t) = \alpha\, U_{\text{buffer}}(t)$$

where $U_{\text{buffer}}(t)$ is the instantaneous step increase in delay in the buffer measured in cells, α is the CBR as a proportion of the maximum available rate on the interface, and $c(t)$ is the transient which can now be analysed in exactly the same way as discussed in the previous section.

8. Conclusions

The analysis contained in this paper suggests that there are some clear issues to be faced in the use of adaptive PLL schemes for CBR services. These include:

- the lower the payload bit rate used to carry the ATM cells, the more difficult the problems associated with adaptive schemes, and, while these rates are becoming of increasing practical importance to users of ATM, theoretical analysis has centred on ATM cells in VC-4,
- even without cell delay variation from ATM multiplexing, waiting time jitter will cause significant amounts of jitter under certain conditions,
- if an adaptive PLL desynchroniser is to be able to adequately attenuate waiting time jitter peaks, then it must have a pole of around 1 mHz,
- in order to maintain closed loop stability, the pole of frequency predictor part of the PLL must be around ten times lower than that of the main loop, i.e. around 0.1 mHz,
- without special design (which would almost certainly be necessary anyway) the acquisition time of the PLL will be around 10 000 sec and, even with special acquisition circuitry, will be significant,

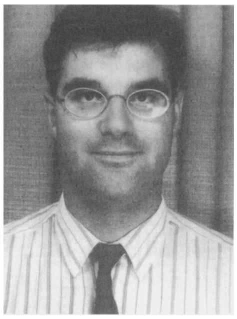
- the level of jitter from multiplexing CDV is most likely to be controlled by the changes in loading on the multiplex and not as a result of 'steady state' conditions and these are likely to produce phase transients which are significantly higher than those of waiting time jitter,

- wander from waiting time jitter and CDV cannot be eliminated by an adaptive PLL desynchroniser as it will be tracked and passed out.

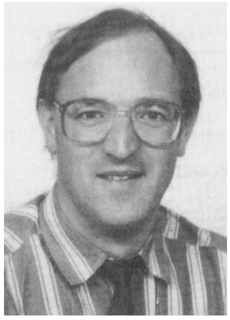
While these issues are not insurmountable, they do present a very stringent design challenge. By comparison, the mechanisms which are based on a network clock, i.e. network synchronous and SRTS, appear by comparison to be simple and robust.

Reference

1 Mulvey M, Kim Y I and Reid A B M: 'Timing issues of constant bit rate services over ATM', BT Technol J, 13, pp 35—45 (July 1995).



Andy Reid joined BT in 1982 after gaining a BSc in electrical and electronic engineering from Dundee University. He initially worked on the design acceptance and integration of TMUX equipment into BT's international network and then was responsible for the product management of international KiloStream. He then joined the team representing BT's transmission interests in ITU-T, ETSI, and ANSI/TI and worked on most aspects of the SDH standards including significant contributions to G.803. He is now responsible for BT's transmission network strategy.



Mick Mulvey received a BEng degree in Electronics from Liverpool University in 1981 and an MSc degree in Telecommunications and Information Systems from Essex University in 1992.

He has worked for BT since 1981 and was originally involved in the development of undersea systems and networks. More recently he has been involved in the development of ATM networks.